might not contain traces of Soviet or hydrogen atomic bomb tests, and return, all within approximately seven hours. There was one inherent hazard in this otherwise routine flight: because the magnetic North Pole is not coterminous with the geographic one, magnetic compasses cannot be used for orientation, forcing pilots to navigate by the stars. The only things the pilots could have with him were a sextant and star charts. They were also instructed to maintain radio silence. The U-2, a product of the latest twentieth-century technology, had to be navigated with the help of an eighteenth-century instrument on these flights.

Maultsby took off for the North Pole from Eielson Air Force Base near Fairbanks, Alaska, around midnight local time on Friday, October 26, with almost all the flight taking place in the early hours of October 27. The pilot's last electronic contact with earth occurred approximately an hour after takeoff, when he picked up a signal from the radio beacon on Barter Island off the northern coast of Alaska. That was also the point at which the pilots of the search and rescue planes that had accompanied him bade him farewell.

At first the unaccompanied flight went without a hitch. Maultsby checked his location and his route with the help of charts and sextant, but as he got closer to the North Pole, he had trouble orienting the sextant toward major stars indicated on the charts because his vision was blurred by flashes of colored lights in the sky. He was looking at the aurora borealis, or northern lights, a light phenomenon caused by the knotting together of a number of the sun's electromagnetic fields. Maultsby could not get his bearings by using the charts and sextant. He decided to fly toward what he believed to be the North Pole, collect the air samples in the filter-paper device and bottles designed for the purpose, and then turn back to fly home. But because the North Pole did not lie where Maultsby believed it to be, he got lost and could not find his way back.

The realization that he was lost came to Maultsby only after he had been flying for eight hours and his fuel supply was almost exhausted. He expected to pick up a radio signal from the Alaskan island he had passed on his way to the North Pole, but there was none. Instead, he heard the voice of a rescue pilot asking him to identify his location by the stars. Maultsby saw Orion 15 degrees to the left of his plane's nose. The rescue pilot who was supposed to meet him over Barter Island but could not do so suggested that